

Introduction to Data Lineage Analysis - Building a Data Flow - ETL (Extract, Transform, Load) Process - Usage of Data Warehouse - Edge Intelligence in Data Flow - Understanding Data Lineage - How Data Lineage Works - Benefits of Data Lineage - Data Lineage Tool Features.

Reference Books:

1. Charles M.Judd, Data Analysis: A Model Comparison Approach To Regression, ANOVA, and Beyond 3rd Edition - By Routledge (2017)
2. Pierre-Yves Bonnefoy, Emeric Chaize, Raphaël Mansuy & Mehdi TAZI, The Definitive Guide to Data Integration 1st Edition - By Packt Publishing (2024).

Title	Generative Adversarial Networks (Professional Elective – I)	Code	CS60507
Prerequisite		Credits Total Hours	3-0-0 (3) 45

Course Outcome:

Students gain comprehensive understanding of deep learning techniques and generative AI models.

CO1: Understand generative models such as generative adversarial networks (GANs) and their advanced techniques.

CO2: To build sophisticated and robust GAN models using PyTorch & convolutional layers etc.,

CO3: Student to learn about the advantages and disadvantages of different GAN performance measures.

CO4: Students to explore and examine the applications of GANs

Unit 1: Basic Generative Adversarial Networks (GANs) 9

Overview of GenAI - Intro to GANs - Learn about GANs and their applications, understand the intuition behind the basic components of GANs -build your very own GAN using PyTorch - Deep Convolutional GAN - Build a more sophisticated GAN using convolutional layers - Learn about useful activation functions - batch normalization - and transposed convolutions to tune your GAN architecture and apply them to build an advanced DCGAN specifically for processing images

Unit 2: Build Basic Generative Adversarial Networks 9

Wasserstein GANs with Normalization - Reduce instances of GANs failure due to imbalances between the generator and discriminator by learning advanced techniques such as WGANs to mitigate unstable training and mode collapse with a W-Loss and an understanding of Lipschitz Continuity - Conditional and Controllable GANs - Understand how to effectively control your GAN - modify the features in a generated image - and build conditional GANs capable of generating examples from determined categories.

Unit 3: Build Better Generative Adversarial Networks (GANs) 9

GAN Evaluation - Understand the challenges of evaluating GANs - learn about the advantages and disadvantages of different GAN performance measures - and implement the Fréchet Inception Distance FID method using embeddings to assess the accuracy of GANs -GAN Disadvantages and Bias - Find out the disadvantages of GANs when compared to other generative models - discover the pros/cons of these models — plus - learn about the many places where bias in machine learning can come from - why it's important - and an approach to identify it in GANs - StyleGAN and Advancements - Understand how StyleGAN improves upon previous models and implements the components and the techniques associated with StyleGAN - currently the most state-of-the-art GAN with powerful capabilities.

Unit 4: Apply Generative Adversarial Networks (GANs) 9

GANs for Data Augmentation and Privacy Preservation - Explore the applications of GANs and examine them wrt data augmentation, privacy, and anonymity Improve your downstream AI models with GAN-generated data - Image-to-Image Translation - Leverage the image-to-image translation framework and identify extensions – generalizations - applications of this framework to modalities beyond images -Implement Pix2Pix - a paired image-to-image translation

Unit 5: Apply Generative Adversarial Networks 9

GAN - to adapt satellite images to map routes with advanced U-Net generator -Patch GAN discriminator architectures - Image-to-Image Unpaired Translation - Compare paired image-to-image translation to unpaired image-to-image translation and identify how their key difference necessitates different GAN architectures - Implement CycleGAN- an unpaired image-to-image translation model, to adapt horses to zebras with two GANs in one.

Reference Books:

1. Jakub Langr & Vladimir Bok, GANs in Action: Deep learning with Generative Adversarial Networks - By Manning; 1st edition (2019)
2. John Hany, Hands-On Generative Adversarial Networks with PyTorch 1.x - By Packt Publishing, (2019)

Title	Professional Skills - I	Code	CS69501
Prerequisite		Credits Total Hours	0-0-2 [1] 30

Course Outcome:

Students to understand professional, behavioral and presentation skills while working with team and practically experience important aspects of it.

CO1: To help the students understand and implement positive outlook, interpret the body language of team members and stakeholders, better interpersonal relationships. Develop into self-motivated professionals with confidence. Practice Responding instead of Reacting.

CO2: Create good Presentation and Present with confidence. Also, recognize and manage Stress, Prioritize and Plan.

CO3: To listen to understand. To be able to ask good questions.

CO4: To understand to be a good Team player, Team Dynamics and to understand the Business Ethics

CO5: To be able to write and speak correctly, forming grammatically correct sentences.

Unit 1: Positive Attitude

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Attitude- Campus to Corporate attitude change, Recognizing Negative Attitude, Campus to Corporate attitude change; Attitude at work- Impact of Negative Attitude in the Workplace, Overcoming Negative Attitude, positive attitude, thought process, Building self-confidence and Assertiveness; Toxic positivity; 3Es, Motivation-Intrinsic and Extrinsic Motivation, Inspiration vs motivation; Emotional Intelligence-Intro to EI, Four clusters. Transactional Analysis (TA), SWOT analysis - Professional analysis

Unit 2: Body Language

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Importance of Body Language, Five Cs of Body Language, Body language in different cultures, Positive Body Language; Voice Control- Pace. Pause and Pitch; Culture-Inclusivity and Proxemics across Global Cultures, Understanding POSH; Stress Management-What is Stress, Eustress, Reasons of stress (work/ personal); Stress